Who can challenge, correct and decide?

Selected AI-governance commitments in Magnifica Humanitas, explained and checked

Travis Darshan | November 3, 2026

Research, formalization and drafting supported by ChatGPT (OpenAI).

Release DOI: 10.5281/zenodo.23923506

Key findings from the Lean models

- A named help desk may still leave the person responsible for your case unidentified.
- Internal review, your right to appeal and the power to correct a decision are different protections.
- Fixing a scoring rule and fixing an earlier denial require separate actions.
- A review board needs authority over your decision and a way to make its finding effective.
- Recording people's views does not show that those views can change a decision.
- A clear explanation can describe a rule that fails the agreed fairness test.
- A stop must arrive before harm occurs; a corrected record cannot undo every harm.

1. The encyclical and this study

What we did. We studied selected AI-governance commitments in Pope Leo XIV's Magnifica Humanitas. We turned them into precise requirements and checked imagined examples using Lean, software that checks logical proofs. We examined where protections fail and how a changed process can supply them. Our question is practical: does a promised protection reach the person who needs it?

Why this matters. Imagine being denied public assistance. The letter names an agency and offers an appeal. You still need to know who is responsible for your case, whether anyone can change it, and whether a successful review will lead to that change. Our examples separate these questions so an institution can examine each one.

The encyclical in brief. Magnifica Humanitas is a papal teaching letter dated 15 May 2026. Its English PDF has 91 pages and 245 numbered paragraphs. Its five chapters connect Catholic social teaching with technology, truth, work, freedom, power and peace. Babel and Nehemiah frame a choice between domination and shared rebuilding (§§7-10). Human dignity, care for vulnerable people and the common good guide the discussion. Paragraphs 14 and 47 encourage practical application. Our formal study examines selected commitments within §§97-111; the wider text explains why those commitments matter.

What Lean does. Lean checks logical proofs. We write down the rules, the people and the steps in an example. We then prove claims about what can happen, and Lean checks each step of the proof. For instance, a completed review can leave a denial unchanged when nobody applies the finding. We check that failure and a process that completes the change. The assumptions remain visible so readers can examine them.

Source: Leo XIV, Magnifica Humanitas, official English PDF, 15 May 2026. §§7-16 and 47 give context; selected §§97-111 define the formal source scope. The source was accessed on 2 November 2026. Page references throughout use the 91-page English edition.

2. What the selected passages ask of us

Read the source before examining the examples

Why these passages. We chose §§97-111, on pages 33-38 of the official English PDF. They connect responsibility, appeal, oversight, participation and the choices made by developers. Together they let us follow a decision from the rule that produces it to the people who can question and correct it. The following source references explain each connection.

Keep the person in view: §§97-103. These paragraphs discuss human direction and the consequences of AI use. In §§102-103, Pope Leo connects automated exclusion with lost responsibility and the need to recognize that people can change. An applicant needs someone able to hear her circumstances and examine her case. We study specific procedures that can serve that purpose. Compassion and a person's whole life remain wider human concerns.

Make appeal and responsibility real: §§104-105. Paragraph 104 rejects exclusion “without the possibility of appeal”. Paragraph 105 says “responsibility must be clearly defined at every stage”. We examine how an institution can put these commitments into practice. Does it identify the person assigned to this case? Can the affected person submit an appeal? Can someone correct the decision when correction is warranted? Each question needs its own answer.

Give oversight and participation an effect: §§106-109. These passages call for institutional safeguards and for communities to help shape choices and corrections. We examine what a board may review and what people can make happen through their input. For example, a request can be recorded yet have no effect on whether a disputed action is paused. Our comparison asks what changes when the process gives that request an effect.

Distinguish power from authority: §§110-111. Paragraph 110 challenges the idea that “technical power automatically confers the right to govern”. Paragraph 111 addresses developers' responsibilities. We therefore ask two separate questions: what can an actor make the system do, and what is that actor authorized to decide? We also distinguish responsibility for designing a rule from responsibility for using it in someone's case.

Source locations: §§97-103, pp. 33-36; §§104-107, p. 36; §108, pp. 36-37; §§109-110, p. 37; §111, pp. 37-38. The three short quotations are exact extracts. The questions and procedures are our interpretation of the cited commitments.

3. From a denial to a corrected record

Follow the applicant through the main benefits example

Start with the denied applicant. The main benefits example uses an assumed eligibility score of 60 and a cutoff of 50. The initial rule subtracts 15 points for assisted applications. That applicant is denied. The study's agreed fairness standard says the way someone applies should not change entitlement. Removing the deduction makes the calculation yield a grant. An earlier denial still needs its own correction.

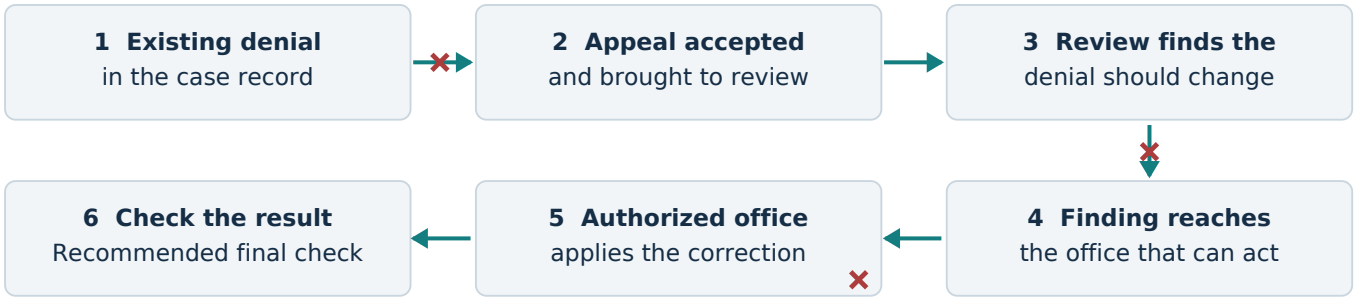
Who is responsible for this case? A general contact can be identifiable while the person assigned responsibility for the denial remains unidentified. Lean checks this gap and a version that identifies the assigned person. Naming the responsible person also leaves a further question: what can that person change?

Follow responsibility when AI is introduced. In the employment example, switching to direct AI production leaves responsibility, appeal and correction protections intact when the institution retains them. Removing the person's assignment breaks the responsibility requirement. A separate comparison shows that a developer assigned to design review does not fill an unassigned role for a later denial. The institution must provide that later assignment too.

Let the person bring a challenge. An office can review a rejection internally while accepting no appeal from the rejected applicant. Lean checks that example. We also distinguish a channel that reviews a challenge from one with power to correct the decision. Opening appeal earlier does not itself create correction power. These are separate steps in protecting the affected person.

Carry the finding through to a change. In the benefits example, review under the corrected rule finds that the stored denial should become a grant. The denial remains if the finding never reaches the office that can act, or if nobody applies the change. Lean checks the complete route shown below, including the required order. The changed record can then be checked. Payment requires a separate action.

A named person remains responsible for the case throughout



Red gaps: appeal not accepted; finding not delivered; office does not apply the correction.
In this example, any of these gaps leaves the stored denial unchanged.

Source connections: §§103-106, 108 and 111, pp. 35-38. Technical evidence: §§3-5 (appeal, responsibility and disclosure), §8 (design and use), §11 (the completed correction route). The diagram shows the main benefits example when review finds a correction is needed.

4. Who can review, change and stop a decision?

Give each person or body a clearly defined job and power

Authorize the board to examine this decision. A board may review general inquiries while lacking authority over a denial. Another board may cover the denial but have no way to intervene. Lean checks both gaps and versions that supply the missing powers. In the appeal example, the board's useful power is a route through an office that can apply its finding.

Show what people's input can change. Residents ask to pause a disputed denial. At first, their request is recorded but never causes a pause. Enabling their request changes that result while preserving the operator's and board's powers to stop the action. Another example separates advice that influences an outcome from control over the final choice. The process should make clear what each form of participation can do.

Look behind an accurate explanation. Return to the two applicants with assumed scores of 60. Subtracting 15 points for assisted intake produces one denial. The notice can explain that calculation accurately while the rule fails our agreed fairness standard. Someone might need assistance because of disability, language or limited connectivity. Those are illustrations of why the issue matters, rather than additional facts in the model.

Make the revised rule reach the calculation. Recording an objection leaves the deduction unchanged. In the checked repair, the assigned design reviewer has authority to change the rule, and the proposed change passes the stated assessment. The calculator then uses the corrected rule. Lean also proves that it survives every finite sequence of the permitted revision attempts. This guarantee concerns the two candidate rules and fixed checks in the example.

Ask who authorized the action. An operator can technically make a review rule produce either outcome while lacking a recognized mandate for that decision. Recognizing the mandate changes authorization without changing technical ability. Our authority test requires that recognition and the right scope. This makes the distinction explicit: an ability to control the system does not supply those institutional permissions.

Source connections: §§104-111, pp. 36-38. Technical evidence: §6 (board powers), §7 (participation and pause), §8 (authority), §9 (the fairness test), §10 (changing the rule).

5. What the checks show

Timing, different requirements and the limits of the examples

Prevent harm while there is time. Imagine automated equipment about to cause damage. In one checked sequence, a requested stop arrives before the dangerous action. In another, it arrives afterward. Both records eventually show a delivered stop and a corrected log, but only the first avoids damage. Lean checks that later modeled actions cannot reverse the loss. A timely delivered stop also protects a loss-free state through every finite continuation of those actions.

Separate what stays true from what changes. We tested different appeal and correction requirements. Some findings hold across all three policies: earlier appeal access still does not identify an unnamed responsible person. Other findings depend on the chosen requirement, such as access before exclusion or correction power after harm. A repaired benefits example also shows that the selected responsibility, appeal and oversight protections can work together.

What an institution can learn. Take one consequential decision and follow it. Who is responsible? Can the person challenge it? Can the reviewer examine this case? Who receives the finding and applies a warranted correction? The answers can reveal a missing connection. They also tell an institution what evidence it needs to show that a protection works.

How the evidence fits together. Each result connects a source passage, our interpretation, stated assumptions and a checked conclusion. The technical paper gives the exact statements and proof references. It also compares the findings with selected EU human-oversight and automated-decision provisions (§15). That legal discussion explains related questions; it adds no new legal premises to the Lean models.

How to read the examples. The scores, roles, permissions and actions are stated assumptions in imagined cases. Lean checks what follows from them. The study covers selected commitments within §§97-111, not the whole encyclical. A real institution must provide evidence of its own procedures, accessibility and results. The examples do not certify it. The technical paper states the assumptions for each result.

Technical evidence: §12 (timing), §13 (comparison of requirements), §14 (result types and proof audit), §15 (legal comparison). The formal proofs remain at checkpoint 0.16. A correction to the record and delivery of support are separate outcomes.

6. Put the findings to work

A six-step guide for examining one institutional decision

Examine one complete case. Use the table to follow a decision from its governing rule to an implemented correction. At each stage, ask for evidence of what someone actually did. These are our practical recommendations, informed by the checked examples.

Step	What someone must do	What to check
1. Name responsibility	Identify who is responsible for the rule and for this case.	Named people, their assignments and relevant powers.
2. Examine the rule	Check the rule actually used against the agreed standard.	The current rule and the assessment of it.
3. Accept a challenge	Give the person a usable route to submit an appeal.	The appeal was accepted and reached the reviewer.
4. Deliver the finding	Let an authorized reviewer examine the case and send the finding onward.	The completed finding reached the office that can act.
5. Apply the correction	Have an authorized person change the rule or the case, as needed.	The calculator uses the new rule, or the case record changed.
6. Check the result	Inspect the changed record; track further delivery separately.	The corrected result and any payment or action still due.

Check both the rule and the earlier decision. Removing the 15-point deduction fixes future calculations in our example. It does not change a denial already stored. That case still needs an accepted appeal, completed review, delivery of the finding and an authorized office applying the change. Check the resulting record, then track any payment or other delivery of support separately.

Make the route usable and timely. Check whether the person can use the appeal route in practice, including any assistance needed. Where an action could cause irreversible harm, check whether a stop can reach the system before it acts. These practical checks require evidence from the institution using the system.

The purpose of the work. The aim is a protection people can use: someone answers for the case, a challenge reaches a reviewer, and a warranted correction reaches the decision. This offers institutions a concrete way to examine selected commitments in the encyclical. The final question is whether the promised protection reaches the person who needs it.

Companion evidence: the technical paper, one-page summary and MagnificaHumanitas_V1_Source_and_Reports.zip, release 1.1. The package includes the Lean sources, proof references, assumptions and reproduction instructions. It can be used without repository access. Suggested institutional records in this table are recommendations. Release DOI: 10.5281/zenodo.23923506.

Acknowledgment. Research, formalization and drafting were supported by ChatGPT (OpenAI). Lean 4 was used to check the formal proofs. Travis Darshan is the author and takes responsibility for the interpretations, assumptions and presentation of the work.